Q1 (3) - Single Correct

Here, $(\sqrt{5}-2)$ is a positive proper fraction.

$\therefore (\sqrt{5}-2)^{\rho}$ is also a positive proper fraction say f.

Now, consider $(\sqrt{5}+2)^{p}-(\sqrt{5}-2)^{p}$

$=2\left[{}^{p}C_1(\sqrt{5})^{p-1}\cdot 2+{}^{P}C_3(\sqrt{5})^{p-3}\cdot 2^3+\cdots\right]\{(p-1)\text{ is even}\}$

$1+f-f'=2\left[{}^{p}C_1\cdot(\sqrt{5})^{p-1}\cdot 2+{}^{p}C_3(\sqrt{5})^{p-3}\cdot 2^3+\cdots+{}^{p}C_p\cdot 2^{p}\right]$

Here $-1<f-f'<1$

$\Rightarrow \quad f-f'=0$

$\therefore 1-2^{\rho+1}=2p\left[(\sqrt{5})^{p-1}\cdot 2+\frac{(p-1)(p-2)}{3!}(\sqrt{5})^{\rho-3}\cdot 2^3+\cdots+{}^{p}C_{p-2}\cdot(\sqrt{5})^2\cdot 2^{\rho-2}\right]$

But, $I=\left[(2+\sqrt{5})^{\circ}\right]$

Hence, $\left[(2+\sqrt{5})^{6}\right]-2^{p+1}=2p$ [an integer]

So, must be divisible by $(2p)$.

Q2 (1) - Single Correct

We have,

$\Rightarrow \quad (7)^{1995}\Rightarrow\left(7^{2\times 997}\right)\cdot 7$

$\Rightarrow \quad (49)^{997}\cdot 7\Rightarrow 7(50-1)^{997}$

$\Rightarrow \quad 7\left[50^{997}-{}^{997}C_1 50^{996}+\cdots{}^{997}C_{996}50-(1)^{997}\right]$

$\Rightarrow 7\times 50^2\left[50^{995}-{}^{997}C_1 50^{994}+\cdots-{}^{997}C_{995}\right]+7[997\times 50-1]$

$\Rightarrow \quad 100k+[348950-7]$

$\Rightarrow \quad 100k+348943$

$\Rightarrow \quad 100k+343900+43$

$\Rightarrow \quad 100[k+3489]+43$

$\therefore \quad$ Where, 7^{1995} is divided by 100 leaves the remainder is 43. Hence, option (1) is correct.

Q3 (4) - Single Correct

We have, $(x+1)(x+2)(x+3)\ldots(x+10)$

$+(x+2)(x+3)\ldots(x+11)+\ldots+(x+11)(x+12)\ldots(x+20)$

Coefficient of x^9 in the given polynomial is

$(1+2+3+\cdots+10)+(2+3+4+\cdots+11)\cdots(11+12+13+\cdots+20)$

$55+65+75+\cdots+155$

$=\frac{11}{2}(55+155)=\frac{11}{2}\times 210=1155$

Hence, option (4) is correct.

Q4 (3) - Single Correct

We have, $(1+x)^n(1+y)^n(1+z)^n$

$=\left(\sum_{p=0}^{n}{}^nC_p x^p\right)\left(\sum_{q=0}^{n}{}^nC_q y^q\right)\left(\sum_{s=0}^{n}{}^nC_s z^s\right)$

$=\sum_{0\le p,q,s\le n}\left({}^nC_p\right)\left({}^nC_q\right)\left({}^nC_s\right)x^p y^q z^s$

For the coefficient of degree r, we must have $p+q+s=r$ where, p,q,s are integers with $\rho,q,s\geq 0$

Sum of such coefficient $=\sum_{p,q.s\ge 0}\left({}^nC_p\right)\left({}^nC_q\right)\left({}^nC_s\right)={}^{3n}C_r$

$[\because p+q+s=r]$

Hence, option (3) is correct.

Q5 (3) - Single Correct

We have, $\sum_{r=0}^{n}(-1)^r\left(\frac{{}^nC_r}{{}^{r+3}C_r}\right)=\sum_{r=0}^{n}(-1)^r\frac{n!(r!)(r+3-r)!}{r!(n-r)!(r+3)!}$

$=3!\sum_{r=0}^{n}(-1)^r\frac{n!}{(n-r)!(r+3)!}$

$=\frac{3!}{(n+1)(n+2)(n+3)}\sum_{r=0}^{n}(-1)^r\frac{(n+3)!}{(n-r)!(r+3)!}$

$=\frac{3!}{(n+1)(n+2)(n+3)}\sum_{r=0}^{n}(-1)^{r n+3}C_{r+3}$

$=\frac{3!(-1)^3}{(n+1)(n+2)(n+3)}\sum_{s=3}^{n+3}(-1)^{s n+3}C_s$

$=\frac{-3!}{(n+1)(n+2)(n+3)}$

$\left[\left(\sum_{s=0}^{n+3}(-1)^s n+3C_s\right)-\left({}^{n+3}C_0-n+3C_1+n+3C_2\right)\right]$

$=\frac{3!}{(n+1)(n+2)(n+3)}\times\left[-(n+2)+\frac{(n+3)(n+2)}{2!}\right]$

$=\frac{3!}{(n+1)(n+2)(n+3)}\left[\frac{(n+1)(n+2)}{2}\right]=\frac{3!}{2(n+3)}=\frac{3}{(n+3)}$

Hence, option (3) is correct.

Q6 (3) - Single Correct

We have,

$\frac{3^{2n}}{8} = \frac{9^n}{8} = \frac{1}{8}(8+1)^n = \frac{1}{8}\left[8^n + {}^nC_1 8^{n-1} + \cdots + {}^nC_{n-1}8 + 1\right]$

$= \frac{1}{8} \times 8\left[8^{n-1} + {}^nC_1 8^{n-2} + \cdots + {}^nC_{n-1}\right] + \frac{1}{8}$

$= \left[8^{n-1} + {}^nC_1 8^{n-2} + \cdots + {}^nC_{n-1}\right] + \frac{1}{8} = k + \frac{1}{8}$[$k$ is an integer]

$\therefore$ Fractional part of $\frac{3^{2n}}{8}$ is $\frac{1}{8}$.

Hence, option (3) is correct.

Q7 (1) - Single Correct

We have, $13^n + 7^n - 3^n = 13^n - 3^n + 7^n = 10k + 7^n$

$[\because 13^n - 3^n$ is divisible by $13 - 3 = 10]$

$= 10k + \left(7^4\right)^{\left(\frac{n}{4} - \frac{1}{4}\right)} \cdot 7 = 10k + (2401)^{\frac{n-1}{4}} \cdot 7$

$= 10k + (2400 + 1)^{\frac{n-1}{4}} \cdot 7 = 10k + 100k + 7$

The unit digit is 7, if $\frac{n-1}{4} = k$

$n = 4k + 1, k \in I$

Hence, option (1) is correct.

Q8 (4) - Single Correct

We have, $S = \sum_{r=0}^{n-1} \frac{1}{(n-r)^2}\left(\frac{{}^nC_{r+1}}{{}^nC_r}\right)^2$

$S = \sum_{r=0}^{n-1} \frac{1}{(n-r)^2}\left[\frac{n-r}{r+1}\right]^2 \left[\because \frac{{}^nC_{r+1}}{{}^nC_r} = \frac{n-r}{r+1}\right]$

$S = \sum_{r=0}^{n-1} \frac{1}{(r+1)^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2}$

Hence, option (4) is correct.

Q9 (3) - Single Correct

Let coefficient of y^m in $e^{xy}(e^y - 1)^m$

$=$ Coefficient of y^m in $e^{xy}\left\{e^{my} - {}^mC_1 e^{(m-1)y} + \cdots + (-1)^m \cdot {}^mC_m\right\}$

$\Rightarrow$ Coefficient of y^m in $e^{xy}\left\{\left(1 + \frac{y}{1!} + \frac{y^2}{2!} + \cdots\right) - 1\right\}^m$

$=$ Coefficient of y^m in $\left\{e^{xy}\left[e^{my} - {}^mC_1 e^{(m-1)y} + \cdots + (-1)^m \cdot {}^mC_m\right]\right\}$

$=$ Coefficient of y^m in $e^{xy} y^m \left(1 + \frac{y}{2!} + \frac{y^2}{3!} + \cdots\right)^m$

$=$ Coefficient of y^m in

$\{e^{(x+m)y} - {}^mC_1 \cdot e^{y(m+x-1)} + \cdots + (-1)^m \cdot {}^mC_m \cdot e^{xy}\}$

$\Rightarrow \quad 1 = \frac{(x+m)^m}{m!} - {}^mC_1 \cdot \frac{(x+m-1)^m}{m!} + \cdots + (-1)^m \frac{{}^mC_m \cdot x^m}{m!}$

$\Rightarrow \quad m! = (x+m)^m - {}^m \quad C_1 \cdot (x+m-1)^m + {}^mC_2(x+m-2)^m + \cdots + (-1)^m \cdot {}^mC_m$

Q10 (2) - Single Correct

$(2010)^n - 2(1010)^n - 4(510)^n + 5(10)^n$

$\Rightarrow (2000+10)^n - 2(1000+10)^n - 4(500+10)^n + 5(10)^n$

Here, $(2000+10)^n = {}^nC_0(2000)^n + {}^nC_1(2000)^{n-1} \cdot (10) + \cdots$

$+{}^nC_{n-1}(2000)(10)^{n-1} + {}^nC_n 10^n$

$=$ Except last term the remaining terms are divisible by 10^3.

$= N_1 10^3 + 10^n$ ...(i)

$$\text{Similarly,} \quad 2(1010)^n = 2N_2 \cdot 10^3 + 2(10)^n \quad \ldots \text{(ii)}$$
$$4(510)^n = N_3 \cdot 10^3 + 4(10)^n \quad \ldots \text{(iii)}$$

From Eqs. (i), (ii) and (iii), we get

$$= (2010)^n - 2(1010)^n - 4(510)^n + 5(10)^n$$
$$\Rightarrow N_1 \cdot 10^3 + 10^n - 2N_2 \cdot 10^3 - 2(10)^n$$
$$-N_3 \cdot (10)^3 - 4(10)^n + 5(10)^n$$

$\Rightarrow 10^3 (N_1 - 2N_2 - N_3)$ where, $N_1, N_2, N_3 \in N$

$\therefore$ The given number is always divisible by 10^3

Q11 (1,3,4) - Multiple Correct

We have. $(9+\sqrt{80})^n = I + f$

Clearly. $0 < 9 - \sqrt{80} < 1$

$\therefore 0 < (9-\sqrt{80})^n < 1$

Let $(9-\sqrt{80})^n = F, \quad 0 < F < 1$

Now $I + f = (9+\sqrt{80})^n$

$= 9^n + {}^nC_1 9^{n-1}\sqrt{80} + {}^nC_2 9^{n-2}(\sqrt{80})^2 + \cdots + (\sqrt{80})^n \ldots (i)$

and $F = (9-\sqrt{80})^n$

$$= 9^n - {}^nC_1 9^{n-1}\sqrt{80} + {}^nC_2 9^{n-2}(\sqrt{80})^2 \cdots (-1)^n \sqrt{80}^n \ldots .\text{(ii)}$$

On adding Eqs. (i) and (ii), we get

$$\begin{aligned} I+f+F &= 2\left[9^n + {}^nC_2 9^{n-2}(\sqrt{80})^2 + \cdots\right] \\ &= 2\left[9^n + {}^nC_2 9^{n-2}\cdot 80 + \cdots\right] \end{aligned}$$

$\therefore \quad I+f+F=$ an even number

$\Rightarrow \quad f+F=$ an even number $-1=$ an integer

$\because \quad 0<f<1$

$\therefore \quad 0<t+F<2$

$\therefore \quad f+F=1 \quad \Rightarrow F=1-f=(9-\sqrt{80})^n$

$1+1=$ an even number

$\therefore \quad I=$ an even number $-1=$ an odd number

$$\begin{aligned} (I+f)(I-f) &= (I+f)(F) \\ &= (9+\sqrt{80})^n(9-\sqrt{80})^n = (81-80)^n = 1 \end{aligned}$$

Hence, option (1), (3) and (4) are correct.

Q12 (1,3) - Multiple Correct

We have,

${}^{100}\mathrm{C}_6 + 4\cdot{}^{100}\mathrm{C}_7 + 6\cdot{}^{100}\mathrm{C}_8 + 4\cdot{}^{100}\mathrm{C}_9 + {}^{100}\mathrm{C}_{10}$

$\Rightarrow {}^4\mathrm{C}_0{}^{100}\mathrm{C}_{94} + {}^4\mathrm{C}_1{}^{100}\mathrm{C}_{93} + {}^4\mathrm{C}_2{}^{100}\mathrm{C}_{92} + {}^4\mathrm{C}_3{}^{100}\mathrm{C}_{91} + {}^4\mathrm{C}_4{}^{100}\mathrm{C}_{90}$

$(1+x)^4 = {}^4C_0 + {}^4C_1x + {}^4C_2x^2 + {}^4C_3x^3 + {}^4C_4x^4$

and $(1+x)^{100} = {}^{100}C_0 + {}^{100}C_1x^2 + {}^{100}C_3x^3$

$\cdots + {}^{100}C_{94}x^{94} + {}^{100}C_{95}x^{95} + \cdots + x^{100}]$

$(1+x)^4(1+x)^{100} = {}^4C_0{}^{100}C_0 + {}^4C_1{}^{100}C_1x^3 + \cdots$

$(1+x)^{104} = \left[{}^4C_0{}^{100}C_0 + {}^4C_1{}^{100}C_1x^3 + \cdots\right]$

Coefficient of x^{94}, we get

${}^{104}C_{94} = {}^4C_0{}^{100}C_{94} + {}^4C_1{}^{100}C_{93} + {}^4C_2{}^{100}C_{92} + {}^4C_3{}^{100}C_{91} + {}^4C_4{}^{100}C_{90}$

$\therefore \quad x+y=104+94=198$

Also, ${}^{100}C_6 + 4\cdot{}^{100}C_7 + 6\cdot{}^{100}C_8 + 4\cdot{}^{100}C_9 + {}^{100}C_{10}$

$\Rightarrow {}^4C_4{}^{100}C_6 + {}^4C_3{}^{100}C_7 + {}^4C_2{}^{100}C_8 + {}^4C_1{}^{100}C_9 + {}^4C_{10}{}^{100}C_{10}$

Coefficient of x^{10} in $(1+x)^{104}$ is

${}^{104}C_{10} = {}^4C_4{}^{100}C_6 + {}^4C_3{}^{100}C_7 + {}^4C_2{}^{100}C_8 + {}^4C_1{}^{100}C_9 + {}^4C_{10}{}^{100}C_{10}$

$\therefore \quad x+y=104+10=114$

Hence, option (1) and (3) is correct.

Q13 (2,3) - Multiple Correct

We have, $\left(x^2+\frac{1}{x^2}-2\right)^{10} \Rightarrow \left(x-\frac{1}{x}\right)^{20}$

$T_{r+1}={}^{20}C_r(x)^{20-r}(-x)^{-r}$

$T_{r+1}={}^{20}C_r(x)^{20-2r}(-1)^r$

The value of constant term in the binomial expansion $\left(x-\frac{1}{x}\right)$

$20-2r=0 \Rightarrow r=10$

$T_{10+1}={}^{20}C_{10}(-1)^{10}={}^{20}C_{10}$

(1) Number of different dissimilar term in the expansion

$(x_1+x_2+x_3+\cdots+x_{10})^{10}={}^{10+10-1}C_{10-1}={}^{19}C_9$

(2) $\left({}^{10}C_0\right)^2+\left({}^{10}C_1\right)^2+\cdots+\left({}^{10}C_{10}\right)^2=\frac{(2\times10)!}{10!10!}=\frac{20!}{10!10!}={}^{2Q}C_{10}$

$\left[\because ({}^nC_1)^2+({}^nC_2)^2+\cdots+({}^nC_n)^2=\frac{2n!}{n!n}\right]$

(3) Coefficient of x^{10}, in $(1-x)^{20}={}^{20}C_{10}(-1)^{10}={}^{20}C_{10}$

(4) Number of linear arrangement of 20 things of which 10 alike one kind and rest all are different, taken all at a time $=\frac{20!}{10!}$

Hence, option (2) and (3) is correct.

Q14 (1,4) - Multiple Correct

We have, $\sum_{k=0}^{100}{}^{100}C_k(x-2)^{100-k}3^k$

$(x-2+3)^{100}=\sum_{k=0}^{100}100C_k(x-2)^{100-k}3^k$

Coefficient of x^{50} in $(x+1)^{100}={}^{100}C_{50}=\frac{100!}{50!50!}$

(1) Number of ways in which 50 identical books can be distributed in 100 students, if each student can get atmost one book is coefficient of x^{50} in $\left(x^0+x^1\right)^{100}$

ie. coefficient of x^{50} in $(1+x)^{100}={}^{100}C_{50}=\frac{100!}{50!50!}$

(2) Number of ways in which 100 different white balls and 50 identical red balls can be arranged in a circle, if no two red balls are together is 99! $^{99}\mathrm{C}_{50}$

(3) Number of dissimilar terms in $(x_1+x_2+x_3+\cdots+x_{50})^{50}$, is

$50+50-1C_{50-1}={}^{99}C_{49}$

$$(4)\ \frac{2\cdot 6\cdot 10\cdot 14\cdots 198}{50!}=\frac{2^{50}(1\cdot 3\cdot 5\cdot 7\cdots 99)}{50!}$$
$$=\frac{2^{50}[1\cdot 2\cdot 3\cdot 4\cdot 5\cdots 99\cdot 100]}{(2\cdot 4\cdot 6\cdot 8\cdots 100)(50!)}$$
$$=\frac{2^{50}(100!)}{2^{50}[1\cdot 2\cdot 3\cdots 50](50!)}$$
$$=\frac{100!}{50!50!}={}^{100}\mathrm{C}_{50}$$

$\therefore$ Hence, option (1) and (4) is correct.

Q15 (4) - Multiple Correct

$(1-x)^{200}={}^{200}C_0-{}^{200}C_1x+{}^{200}C_2x^2-{}^{200}C_3x^3+\cdots$ ...(i)

and $\left(\frac{1-x^{149}}{1-x}\right)=\left(x^{148}+x^{147}+x^{146}+\cdots+x+1\right)$ ...(ii)

On multiplying and equating coefficient of x^{148}. is

$${}^{200}\mathrm{C}_0-{}^{200}\mathrm{C}_1+{}^{200}\mathrm{C}_2\cdots+{}^{200}\mathrm{C}_{148}$$
$$=\text{ Coefficient of } x^{148} \text{ in }\left[(1-x)^{200}\cdot\frac{(1-x^{149})}{(1-x)}\right]$$
$$=\text{ Coefficient of } x^{148} \text{ in }\left[(1-x)^{199}\left(1-x^{149}\right)\right]$$
$$=\text{ Coefficient of } x^{148} \text{ in } (1-x)^{199}={}^{199}C_{148}\text{ or }{}^{199}\mathrm{C}_{51}$$

Q16 (1,2) - Multiple Correct

4 th term of the given expansion $\left(2+\frac{3x}{8}\right)^{10}$

i.e. $T_4={}^{10}C_3(2)^7\left(\frac{3x}{8}\right)^3$

T_A is the maximum numerical value

$\therefore T_4\geq T_5$

$$\left|\frac{T_4}{T_5}\right|\geq 1\Rightarrow\left|\frac{{}^{10}C_3 2^7\left(\frac{3x}{8}\right)^3}{{}^{10}C_4 2^6\left(\frac{3x}{8}\right)^4}\right|\geq 1$$

$\frac{4}{7}\geq\left|\frac{3x}{16}\right|\Rightarrow|x|\leq\frac{64}{21}$ ...(i)

and $\left|\frac{T_4}{T_3}\right|\geq 1$

$$\Rightarrow\left|\frac{{}^{10}C_3 2^7\left(\frac{3x}{8}\right)^3}{{}^{10}C_2 2^8\left(\frac{3x}{8}\right)^2}\right|\geq 1\Rightarrow\frac{8}{3}\geq\left|\frac{2\times 8}{3x}\right|$$

$|x| \geq 2$ ...(ii)

From Eqs. (i) and (ii), $2 \leq |x| \leq \frac{64}{21}$

Hence, value of x be

$x \in \left[\frac{-64}{21}, -2\right] \cup \left[2, \frac{64}{21}\right]$

Hence, options (1) and (2) are correct.

Q17 (1,2,3,4) - Multiple Correct

We have, $(x+y+z)^{10}$

(1) The total number of terms is ${}^{3+10-1}C_{10} = {}^{12}C_{10} = \frac{12!}{10!2!} = 66$

(2) Coefficient of $x^3y^2z^5$ of the given expansion is $\frac{10!}{3!2!5!}$

(3) The sum of the radical of $x^2y^3z^4$ is $2+3+4=9$ is not equal to 10.

Since, $x^2y^3z^4$ cannot be term of the $(x+y+z)^{10}$

(4) $\because (x+y+z)^{10} = \sum_{\alpha+\beta+\gamma=10} \frac{10!}{\alpha!\beta!\gamma!} x^\alpha y^\beta z^\gamma$

Put $x=y=z=1$ both sides, we have

$\Rightarrow \sum_{\alpha+\beta+\gamma=10} \frac{10!}{\alpha!\beta!\gamma!} = 3^{10}$

Hence, (1), (2), (3) and (4) are correct

Q18 (1,2,4) - Multiple Correct

(1) We have $\frac{(1+2x)^n+(1+3x)^n}{1-x}$

$\Rightarrow (1+2x)^n(1-x)^{-1} + (1+3x)^n(1-x)^{-1}$

$\Rightarrow \left[1 + {}^nC_1(2x) + {}^nC_2(2x)^2 + \cdots\right]\left[1+x+x^2+\cdots\right]$

$+ \left[1 + {}^nC_1(3x) + {}^nC_2(3x)^2 + \cdots\right]\left[1+x+x^2+\cdots\right]$

Coefficient of x^n is

$\left(1 + {}^nC_1 2 + {}^nC_2 2^2 + {}^nC_3 2^3 + \cdots + {}^nC_n 2^n\right) + \left(1 + {}^nC_1 3 + {}^nC_2 3^2 + \cdots + {}^n, 3^n\right)$

$\Rightarrow \quad (1+2)^n + (1+3)^n = 3^n + 4^n$

(1) is true.

(2) We have, $\sum_{i=0}^{n}(-1)^r {}^nC_r\left[\frac{1}{2'} + \frac{3^r}{2^{2r}} + \frac{7^t}{2^{3r}} + \cdots \infty\right.$

$$\Rightarrow \sum_{r=0}^{n}(-1)^r {}^nC_r\left(\frac{1}{2}\right)^r + \sum_{r=0}^{n}(-1)^r {}^nC_r\left(\frac{3}{4}\right)^f + \sum_{f=0}^{n}(-1)' C_f\left(\frac{7}{8}\right)^r + \ldots$$

$\Rightarrow \left(1-\frac{1}{2}\right)^n+\left(1-\frac{3}{4}\right)^n+\left(1-\frac{7}{8}\right)^n+\cdots\infty$

$\Rightarrow \left(\frac{1}{2}\right)^n+\left(\frac{1}{4}\right)^n+\left(\frac{1}{8}\right)^n+\cdots\left[\because \sum_{t=0}^{n}(-1)^{rn}C, x^r=(1-x)^n\right]$

$\Rightarrow \frac{1}{2^n}+\frac{1}{2^{2n}}+\frac{1}{2^{3n}}+\cdots \Rightarrow \frac{\frac{1}{2^n}}{1-\frac{1}{2^n}}=\frac{1}{2^n-1}$

Hence, (2) is true.

(3) We have, $\left(1+x+x^2\right)^{10} \Rightarrow \left(\frac{1-x^3}{1-x}\right)^{10}$

$\Rightarrow \quad \left(1-x^3\right)^{10}(1-x)^{-10}$

$\Rightarrow \quad \left(1-{}^{10}C_1x^3+{}^{10}C_2x^6+\cdots\right)(1-x)^{-10}$

Coefficient of x^2 is depend on $(1-x)^{-10}$

$\therefore$ Coefficient of x^2 of $(1-x)^{-10}$ is

${}^{2+10-1}C_{10-1}={}^{11}C_9$

${}^{11}C_9 \neq {}^{12}C_{10}$

Hence, (3) is false.

(4) Coefficient of x^2 in $\left(1+x+x^2\right)^{10}$ is ${}^{11}C_9={}^{11}C_2$

$\therefore (d)$ is true.

Q19 (1,2,3,4) - Multiple Correct

Given,

$1+(1+x)+(1+x)^2+\cdots+(1+x)^n=\sum_{k=0}^{n} a_k x^*$

$\Rightarrow \frac{\left[(1+x)^{n+1}-1\right]}{1+x-1}=\sum_{k=0}^{n} a_k x^k$

$\left[\because 1+r+r^2+r^3+\cdots+r^n=\frac{r^{n+1}-1}{r-1}\right]$

$\Rightarrow \frac{\left[(1+x)^{n+1}-1\right]}{x}=\sum_{k=0}^{n} a_k x^k$ ...(i)

Put $x=1$ in Eq. (i), we get

$\Rightarrow 2^{n+1}-1=\sum_{k=0}^{n} a_k$

Hence, option (1) is correct.

Also, coefficient of $x^k \Rightarrow a_k={}^{n+1}C_{k+1}$

$a_{n-2}={}^{n+1}C_{n-1}=\frac{n(n+1)}{2}$

$\Rightarrow$ (2) is correct.

Also, $a_p>a_{p-1} \Rightarrow p<\frac{n}{2}$

$\therefore$ (3) is correct.

Also, $(a_9)^2-(a_8)^2={}^{n+2}C_{10}\left({}^{n+1}C_{10}-{}^{n+1}C_9\right)$

$\Rightarrow$ (4) is correct.

Q20 (1,2,4) - Multiple Correct

Given,

$$f(m)=\sum_{i=0}^{m}{}^{30}C_{30-i}{}^{20}C_{m-i}$$
$$=\sum_{i=0}^{m}\binom{30}{30-i}\binom{20}{m-i}$$

$={}^{30}C_{30}{}^{20}\mathrm{C}_m+{}^{30}\mathrm{C}_{29}{}^{20}\mathrm{C}_{m-1}+\cdots+{}^{30}\mathrm{C}_{30-m}\times{}^{20}\mathrm{C}_0$

$={}^{30}C_0{}^{20}C_m+{}^{30}C_1{}^{20}C_{m-1}+\cdots+{}^{30}C_m{}^{20}C_0\quad\left[\because{}^{n}C_r={}^{2}C_{n-r}\right]$

$=$ Coefficient of x^m in the expansion of product $(1+x)^{30}(x+1)^{20}$

$=$ Coefficient of x^m in the expansion of $(1+x)^{50}={}^{50}C_m$ Now, ${}^{50}C_m$ will be maximum when $m=25$

Also, $f(0)+f(1)+f(2)+\cdots+f(50)$

$={}^{50}\mathrm{C}_0+{}^{50}\mathrm{C}_1+{}^{50}\mathrm{C}_2+\cdots+{}^{50}\mathrm{C}_{50}=2^{50}\quad\left[\therefore\sum_{r=0}^{n}C_r=2^n\right]$

Also, ${}^{50}\mathrm{C}_m$ is not divisible by 50 for any m as 50 is not a prime number.

$\sum_{i=0}^{m}[f(m)]^2=\left({}^{50}C_0\right)^2+\left({}^{50}C_1\right)^2+\left({}^{50}C_2\right)^2+\cdots+\left({}^{50}C_{50}\right)^2={}^{100}C_{50}$

$\left[\because C_0^2+C_1^2+C_2^2+\cdots+C_n^2={}^{2n}C_n\right]$

Hence, option (1), (2) and (4) is correct.

Q21 (1,2,3,4) - Multiple Correct

${}^{1000}C_{50}+{}^{999}C_{49}+{}^{998}C_{48}+\cdots+{}^{950}C_0$

$\Rightarrow$ Coefficient of x^{950} in $\left\{(1+x)^{950}+(1+x)^{951}+\cdots+(1+x)^{1000}\right\}$

$\Rightarrow$ Coefficient of x^{950} in

$(1+x)^{950}\cdot\frac{\{(1+x)^{51}-1\}}{(1+x)-1}$

$\Rightarrow$ Coefficient of x^{951} in $\left\{(1+x)^{1001}-(1+x)^{950}\right\}$

$$\Rightarrow\quad{}^{1001}C_{951}={}^{1001}C_{50}={}^{1002}C_{951}-{}^{1001}C_{51}$$
$$={}^{1002}C_{51}-{}^{1001}C_{950}$$

Q22 (1,2,3) - Multiple Correct

We have $a_1 = 2$ and for $n \geq 2$

$a_n = \left(1 + \frac{1}{n}\right)^n = {}^nC_0 + {}^nC_1\left(\frac{1}{n}\right) + {}^nC_2\left(\frac{1}{n}\right)^2 + \ldots\ldots + {}^nC_r\left(\frac{1}{n}\right)^r + \ldots\ldots + {}^nC_n\left(\frac{1}{n}\right)^n$

$= 1 + 1 + \frac{n(n-1)}{2!}\frac{1}{n^2} + \ldots\ldots + \frac{n(n-1)\ldots(n-r+1)}{r!}\frac{1}{n^r} + \ldots\ldots + \frac{n(n-1)\ldots 2.1}{n!}\frac{1}{n^n} \quad \ldots\ldots \text{(i)}$

$= 2 + \frac{1}{2!}\left(1 - \frac{1}{\mathrm{n}}\right) + \frac{1}{3!}\left(1 - \frac{1}{\mathrm{n}}\right)\left(1 - \frac{2}{\mathrm{n}}\right) + \ldots\ldots + \frac{1}{\mathrm{r}!}\left(1 - \frac{1}{\mathrm{n}}\right)\left(1 - \frac{2}{\mathrm{n}}\right)\ldots\left(1 - \frac{\mathrm{r}-1}{\mathrm{n}}\right) + \ldots$

$+ \frac{1}{n!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\ldots\ldots\left(1 - \frac{n-1}{n}\right)$

Hence from (i), $a_n \geq 2$ for all $n \in \mathbb{N}$. Also

$a_n \leq 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \ldots\ldots + \frac{1}{r!} + \ldots\ldots + \frac{1}{n!}$

For $2 \leq r \leq n$, we have $r! = 1.2.3\ldots\ldots r \geq 2^{r-1}$. Thus,

$a_n \leq 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \ldots\ldots + \frac{1}{2^{r-1}} + \ldots. + \frac{1}{2^{n-1}}$

$= 1 + \frac{1-(1/2^n)}{1-(1/2)} = 1 + 2\left(1 - \frac{1}{2^n}\right) = 3 - \frac{1}{2^{n-1}}$

$a_n \leq 3 - \frac{1}{2^{n-1}} < 3 \quad \forall \quad n \geq 1 \quad \Rightarrow a_n < 4 \forall n \geq 1$

Q23 (1,3,4) - Multiple Correct

$I + f = (4 + \sqrt{15})^n$

Let $g = (4 - \sqrt{15})^n$, then $0 < g < 1$

$I + f = {}^nC_0 4^n + {}^nC_1 4^{n-1}\sqrt{15} + {}^nC_2 4^{n-2} 15 + {}^nC_3 4^{n-3}(\sqrt{15})^3 + \ldots\ldots.$

$g = {}^nC_0 4^n - {}^nC_1 4^{n-1}\sqrt{15} + {}^nC_2 \cdot 4^{n-2} \cdot 15 - {}^nC_3 4^{n-3}(\sqrt{15})^3 + \ldots\ldots\ldots$

$\therefore \quad I + f + g = 2\left({}^nC_0 4^n + {}^nC_2 4^{n-2} \cdot 15 + \ldots\ldots\ldots\right) = \text{even integer}$

$\because \quad 0 < f + g < 2 \Rightarrow f + g = 1 \Rightarrow 1 - f = g$

thus I is an odd integer

$1 - f = g = (4 - \sqrt{15})^n$

$(I + f)(1 - f) = (I + f) \cdot g = 1$

Q24 (2) - Comprehension Type

We have, $\left(1 + px + x^2\right)^n = 1 + a_1 x + a_2 x^2 + \cdots + a_{2n} x^{2n}$

Differentiate with respect to x, we get

$n\left(1 + px + x^2\right)^{n-1}(p + 2x) = a_1 + 2a_2 x + 3a_2 x^2 + \cdots + 2na_{2n} x^{2n-1}$

Multiply by $\left(1 + px + x^2\right)$ both side, we get

$n\left(1+px+x^2\right)^n(p+2x)=\left(1+pn+x^2\right)\left(a_1+2a_2x+3a_3x^2\right.$

$\left.+\cdots+2na_{2n}x^{2n-1}\right.$

$n(p+2x)\left(1+a_1x+a_2x^2+\cdots+a_{2n}x^{2n}\right)=\left(1+px+x^2\right)$

$\left(a_1+2a_2x+3a_3x^2+\cdots+(r-1)a_{r-1}x^{r-1}+ra_rx^{r-1}+(r+1)a_{r+1}x^i\right)$

Equating the coefficient of x^r, we get

$$\begin{aligned} nPa_r+2na_{r-1} &= (r+1)a_{r+1}+pra_r+(r-1)a_{r-1} \\ npa_r-pra_r &= (r+1)a_{r+1}+(r-1)a_{f-1}-2na_{f-1} \\ (np-pr)a_r &= (r+1)a_{r+1}+(r-1-2n)a_{r-1} \end{aligned}$$

Hence, option (2) is correct.

Q25 (3) - Comprehension Type

We have, $\left(1+px+x^2\right)^n=1+a_1x+a_2x^2+a_3x^3+\cdots+a_{2n}x^{2n}$

On replacing x by x^4, we get

$\left(1+px^4+x^8\right)^n=1+a_1x^4+a_2x^8+a_3x^{12}+\cdots+a_{2n}x^{8n}$

On dividing by x^3 both side, we get

$\frac{\left(1+px^4+x^8\right)^n}{x^3}=\frac{1}{x^3}+a_1x+a_2x^5+a_3x^9+\cdots+a_{2n}x^{8n-3}$

[On differentiate w.r.t x, we get]

$\frac{nx^3\left(4px^3+8x^7\right)\left(1+px^4+x^8\right)^{n-1}-3x^2\left(1+px^4+x^8\right)^n}{x^6}$

$=\frac{-3}{x^4}+a_1+5a_2x^4+9a_3x^8+\cdots+(8n-3)a_{2n}x^{8n-4}$

put $x=1$ both side, we get

$n(4p+8)(2+p)^{n-1}-3(2+p)^n=-3+a_1$

$+5a_2+9a_3+\cdots+(8n-3)a_{2n}$

$(p+2)^n\cdot(4n-3)+3=a_1+5a_2+9a_3+\cdots+(8n-3)a_{2n}$

when $a_1+5a_2+9a_3+\cdots+(8n-3)a_{2n}$ is divide by $(p+2)$ the

remainder is 3. Hence, option (3) is correct.

Q26 (2) - Comprehension Type

We have, $\left(1+x+x^2\right)^{100}=\sum_{r=0}^{200}a_rx'$

$\Rightarrow \quad \left(1+x+x^2\right)^{100}=a_0+a_1x+a_2x^2+\cdots+a_{200}x^{200}$ ...(i)

put $\frac{1}{x}$ in place of x, we get

$\left(x^2+x+1\right)^{100}=a_0x^{200}+a_1x^{199}+a_2x^{198}+\cdots+a_{200}$ ...(ii)

From Eqs. (i) and (ii), we get

$a_0 = a_{200}, a_1 = a_{199} \cdots a_I = a_{200-t}$

$\therefore a_{56} = a_{200-56} = a_{144}$

Since, $a_{56} = a_{144}$

Hence, option (2) is correct.

Q27 (3) - Comprehension Type

Put $x = 1$ in Eq. (i), we get

$3^{100} = a_0 + a_1 + a_2 + a_3 + \cdots + a_{100} + a_{101} + \cdots + a_{200}$

$3^{100} = 2(a_0 + a_1 + a_2 + \cdots + a_{99}) + a_{100}$

$$[\because a_0 = a_{200}, a_f = a_{200-r}]$$

$$\Rightarrow \quad a_0 + a_1 + a_2 + \cdots + a_{99} = \frac{3^{100} - a_{100}}{2}$$

Hence, option (iii) is correct.

Q28 (4) - Comprehension Type

We have,

$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \cdots + {}^nC_nx^n$

Let $P = \sum_{0 \leq r < s \leq n} \sum_S (r+s)\left({}^nC_r + {}^nC_s\right)$ ...(i)

On replacing r by $(n-r)$ and s by $(n-s)$, then

$P = \sum\sum_{0 \leq r < s \leq n}(n - r + n - s)\left({}^nC_{n-r} + {}^nC_{n-s}\right)$

$P = \sum\sum\sum_{0 \leq r < s \leq n}(2n - r + s)\left({}^nC_r + {}^nC_s\right) \quad \ldots \text{(ii)}$

On adding Eqs. (i) and (ii), we get

$2P = \sum_{0 \leq r < s \leq n} 2n\left({}^nC_r + {}^nC_s\right)$

$P = n\sum\sum_{0 \leq r < s \leq n}\left({}^nC_r + {}^nC_s\right)$

$P = n\left(n2^n\right) = n^2 \cdot 2^n$

Hence, option (4) is correct.

Q29 (2) - Comprehension Type

$\left({}^nC_0+{}^nC_1+{}^nC_2+\cdots+{}^nC_n\right)^2$

$=\sum_{r=0}^n\left({}^nC_r\right)^2+2\sum\sum_{0\leq r<s\leq n}{}^nC_r{}^nC_s$

$\Rightarrow \quad \left(2^n\right)^2={}^{2n}C_n+2\sum\sum_{0\leq r<s\leq n}{}^nC_r{}^nC_s$

$\sum_{0\leq r<s\leq n}{}^nC_r{}^nC_s=2^{2n-1}-\frac{1}{2}{}^{2n}C_n$

We have to find $\sum\sum_{0\leq r<s\leq n}(r+s)\left({}^nC_r+{}^nC_s+{}^nC_r\cdot{}^nC_s\right)$

$=\sum\sum_{0\leq r<s\leq n}(r+s)\left({}^nC_r+{}^nC_s\right)+\sum\sum_{0\leq r<s\leq n}(r+s){}^nC_r{}^nC_s$

$=n^2\cdot 2^n+\left[n\cdot 2^{2n-1}-\frac{n}{2}\cdot{}^{2n}C_n\right]$

$=n^2\cdot 2^n+\frac{n}{2}\left[2^{2n}-{}^{2n}C_n\right]$

Hence, option (2) is correct.

Q30 (1) - Comprehension Type

Put $x=\frac{-1}{x}$ in place of x

$\left(1-\frac{1}{x}+\frac{1}{x^2}\right)^n=a_0-\frac{a_1}{x}+\frac{a_2}{x^2}-\cdots+\frac{a_{2n}}{x^{2n}}$ ...(i)

$\Rightarrow\left(x^2-x+1\right)^n=a_0x^{2n}-a_1x^{2n-1}+a_2x^{2n-2}+\cdots+a_{2n}\ldots$(ii)

On multiplying Eqs. (i) and (ii), we get

$\left(1+x^2+x^4\right)^n=\left(a_0+a_1x+a_2x^2+\cdots+a_{2n}x^{2n}\right)\left(a_0x^{2n}-a_1x^{2n-1}+\cdots+a_{2n}\right)\ldots$....(iii)

On putting x^2 in place of x, we get

$\left(1+x^2+x^4\right)^n=a_0+a_1x^2+a_2x^4+\cdots+a_nx^{2n}+\cdots+a_{2n}x^{4n}$

Coefficient of x^{2n} in $\left(1+x^2+x^4\right)^n$ is a_n

On equating the coefficient of x^{2n}, we get

$a_0^2-a_1^2+a_2^2-a_1^2-\cdots+a_{2n}^2=a_n$

Hence, $k=1$

Q31 (2) - Comprehension Type

We have, $\left(1+x+x^2\right)^n=\sum_{r=0}^{2n}a_rx^r$

On dividing both sides by x^n, we get

$\left(1+x+\frac{1}{x}\right)^n=\sum_{r=0}^{2n}a_rx^{r-n}$

Thus, $a_n=$ constant term in the expansion of $\left(1+x+\frac{1}{x}\right)^n$.

But $\left(1+x+\frac{1}{x}\right)^n={ }^n C_0+{ }^n C_1\left(x+\frac{1}{x}\right)$
$+{ }^n C_2\left(x+\frac{1}{x}\right)^2+\cdots+{ }^n C_n\left(x+\frac{1}{x}\right)^n$

Thus,

$a_n={ }^n C_0+{ }^n C_2\left({ }^2 C_1\right)+{ }^n C_4\left({ }^4 C_2\right)+\cdots$

$a_n=1+a \Rightarrow a_n-a=1$

Hence, option (2) is correct.

Q32 (2) - Comprehension Type

$C_0+C_1 x+C_2 x^2+\cdots+C_n x^n=(1+x)^n$

On differentiate w.r.t. x, we get

$\Rightarrow C_1+2 \cdot C_2 x+3 \cdot C_3 x^2+\cdots+n \cdot C_n x^{n-1}=n(1+x)^{n-1}$

Put, $x=i$

$\left(C_1-3 \cdot C_3+5 \cdot C_5-\cdots\right)+i\left(2 \cdot C_2-4 \cdot C_4+\cdots\right)=n(1+i)^{n-1}$

Taking modulus on both sides and squaring, we get

$\left(C_1-3 C_3+5 C_5 \ldots\right)^2+\left(2 \cdot C_2-4 \cdot C_4 \ldots\right)^2=n^2 \cdot 2^{n-1}$

Q33 (3) - Comprehension Type

We have,

$\frac{{ }^n C_0}{n(n+1)}-\frac{{ }^n C_1}{(n+1)(n+2)}+\frac{{ }^n C_2}{(n+2)(n+3)}-\cdots upto(n+1)$ term

$=\frac{{ }^n C_0}{n(n+1)}-\frac{{ }^n C_1}{(n+1)(n+2)}+$
$\frac{{ }^n C_2}{(n+2)(n+3)}-\cdots+(-1)^n \frac{{ }^n C_n}{2 n(2 n+1)}$

$=\left(\frac{{ }^n C_0}{n}-\frac{{ }^n C_0}{n+1}\right)-\left(\frac{{ }^n C_1}{n+1}-\frac{{ }^n C_1}{n+2}\right)+\cdots(-1)^n\left(\frac{{ }^n C_n}{2 n}-\frac{{ }^n C_n}{2 n+1}\right)$

$=\frac{{ }^n C_0}{n}-\frac{{ }^n C_1}{n+1}+\frac{{ }^n C_2}{n+2}-\cdots(-1)^n \frac{{ }^n C_n}{2 n}-\left[\frac{{ }^n C_0}{n+1}-\frac{{ }^n C_1}{n+2}+\cdots(-1)^n \frac{{ }^n C_n}{2 n+1}\right] \quad \ldots$(i)

Now

$\therefore \quad(1-x)^n={ }^n C_0-{ }^n C_1 x+{ }^n C_2 x^2-\cdots+(-1)^{n n} C_n x^n$

On multiplying both side by x^{n-1}, we get

${ }^n C_0 x^{n-1}-{ }^n C_1 x^n+{ }^n C_2 x^{n+1}-\cdots+(-1)^{n n} C_n x^{2 n-1}=x^{n-1}(1-x)^n$

On integrating both side between the limit 0 to 1, we get

$$\left[{}^nC_0\frac{x^n}{n}-{}^nC_1\ \frac{x^{n+1}}{n+1}+C_2\frac{x^{n+2}}{n+2}-\cdots(-1)^{n n}C_n\frac{x^{2n}}{2n}\right]_0^1$$
$$=\int_0^1 x^{n-1}(1-x)^n dx \quad \ldots\text{(ii)}$$

Using Eq. (ii) in Eq. (i), we have

$$\frac{{}^nC_0}{n(n+1)}-\frac{{}^nC_1}{(n+1)(n+2)}+\cdots(-1)^n\frac{{}^nC_n}{2n(2n+1)}$$
$$=\int_0^1 x^{n-1}(1-x)^n dx-\int_0^1 x^n(1-x)^n dx$$
$$=\int_0^1 x^{n-1}(1-x)^n(1-x)dx$$
$$=\int_0^1 x^{n-1}(1-x)^{n+1}dx$$

Q34 (3) - Integer Type

We have, $\sum_{k=0}^{100}\left(\frac{k}{k+1}\right)^{100}C_k=\frac{a\left(2^{100}\right)+b}{c}$

$\Rightarrow \quad \sum_{k=0}^{100}\left(\frac{k+1-1}{k+1}\right)^{100}C_k=\frac{a(2)^{100}+b}{c}$

$\Rightarrow \quad \sum_{k=0}^{100}\left(1-\frac{1}{k+1}\right){}^{100}C_k=\frac{a\left(2^{100}\right)+b}{c}$

$\sum_{k=0}^{100}{}^{100}C_k-\sum_{k=0}^{100}\frac{{}^{100}C_k}{k+1}=a\frac{\left(2^{100}\right)+b}{c}$

$\Rightarrow \quad 2^{100}-\frac{2^{101}-1}{101}=\frac{a\left(2^{100}\right)+b}{C}\left[\because \sum_{t=0}^{n}\frac{{}^nC_f}{r+1}=\frac{2^{n+1}-1}{n+1}\right]$

$\Rightarrow \quad \frac{101\left(2^{100}\right)-2^{101}+1}{101}=\frac{a\left(2^{100}\right)+b}{c}$

$\Rightarrow \quad \frac{2^{100}(101-2)+1}{101}=\frac{a\left(2^{100}\right)+b}{c}$

$\Rightarrow \frac{99\left(2^{100}\right)+1}{101}=\frac{a\left(2^{100}\right)+b}{c}$

On comparing, we get

$a=99, b=1$ and $c=101$

$\therefore \quad a+b+c=99+1+101=201$

$\therefore$ Number of digits is 3.

Q35 (2) - Integer Type

$\lim_{n\to\infty}\sum_{r=0}^{n}\frac{{}^nC_r}{n^r(r+3)}=\lim_{n\to\infty}\sum_{r=0}^{n}\frac{{}^nC_r}{n^r}\cdot\int_0^1 x^{r+2}dx.$

$=\int_0^1\left(x^2\lim_{x\to\infty}\sum_{i=0}^{n}{}^nC_r\left(\frac{x}{n}\right)^r\right)dx$

$= \int_0^1 \left(x^2 \cdot \lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n\right) dx = \int_0^1 x^2 \cdot e^x \cdot dx = e - 2$

But $\quad e - 2 = e - x$

$\therefore x = 2$

Q36 (3) - Integer Type

We have, $\lim_{x\to 0}\left[\sum_{r=0}^{n}(-1)^{rn}C_r\left(\sum_{k=0}^{n-r}{}^{n-1}C_k \cdot x^k \cdot 2^k\right)\left(x^2-x\right)^r\right]^{1/x} = e^{n\lambda}$

$\Rightarrow \lim_{x\to 0}\left[\sum_{r=0}^{n}(-1)^{rn}C_r(1+2x)^{n-r}\left(x^2-x\right)^t\right]^{1/x}$

$\Rightarrow \quad \lim_{x\to 0}\left[\left(1+2x-x^2+x\right)^n\right]^{1/x} \Rightarrow \lim_{x\to 0}\left(1+3x-x^2\right)^{n/x}$

$\Rightarrow \quad e^{\lim_{x\to 0}\frac{\left(3x-x^2\right)\cdot n}{x}} \Rightarrow e^{3n} \quad \left[\because \lim_{x\to 0}[1+f(x)]^{g(x)} = e^{\lim_{x\to 0} f(x)\cdot g(x)}\right]$

$\Rightarrow \quad e^{3n}$

$\therefore \quad \lambda = 3$

Q37 (9) - Integer Type

${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \cdots + (-1)^r \cdot {}^nC_r$

$\Rightarrow {}^{n-1}C_0 - \left({}^{n-1}C_0 + {}^{n-1}C_1\right) + \left({}^{n-1}C_1 + {}^{n-1}C_2\right)$
$\quad - \left({}^{n-1}C_2 + {}^{n-1}C_3\right) + \cdots + (-1)^r \cdot \left({}^{n-1}C_{r-1} + {}^{n-1}C_r\right)$

$\Rightarrow \quad (-1)^r \cdot {}^{n-1}C_r$

$\therefore \quad (-1)^r \cdot {}^{n-1}C_r = 28$, r must be even

$\Rightarrow \quad {}^{n-1}C_r = 28 = 7 \times 4 = \frac{7\times 8}{2} = {}^8C_2$

$\therefore \quad n - 1 = 8$ and $r = 2$

$\Rightarrow \quad n = 9.$

Q38 (2) - Integer Type

Here,

$$S = \sum_{K=1}^{\infty}\frac{1}{K^2} = 1^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{3}\right)^2 + \cdots +$$

$$t = \sum_{K=1}^{\infty}\frac{(-1)^{\prime K+1}}{K^2} = (1)^2 - \left(\frac{1}{2}\right)^2 + \left(\frac{1}{3}\right)^2 - \cdots$$

$$S-t=2\cdot\left\{\left(\frac{1}{2}\right)^2+\left(\frac{1}{4}\right)^2+\left(\frac{1}{6}\right)^2+\cdots+\right\}=2\cdot\frac{S}{4}=\frac{S}{2}$$

$\Rightarrow \frac{S}{2}=t$

$\therefore \frac{S}{t}=2$

Q39 (81) - Integer Type

We have $17^2=289=290-1$

Now $17^{286}=\left(17^2\right)^{128}=(290-1)^{12s}$

$={}^{128}\mathrm{C}_0(290)^{12\,\mathrm{s}}-{}^{128}\mathrm{C}_1(290)^{127}+{}^{128}\mathrm{C}_2(290)^{126}-\ldots\ldots..$

$-{}^{128}\mathrm{C}_{125}(290)^3+{}^{128}C_{126}(290)^2-{}^{128}\mathrm{C}_{127}(290)+1$

$=1000\text{ m}+\frac{(128)(127)}{2}(290)^2-(128)(290)+1$

where m is a positive integer.

$=1000\text{ m}+(128)(290)[(127)(145)-1]+1$

$=1000\text{ m}+(128)(290)(18414)+1$

$=1000\text{ m}+683527680+1$

$=1000[\text{ m}+683527]+680+1$

$=1000[\text{ m}+683527]+681$

Thus, the last two digits are 81.

Q40 (2) - Integer Type

$x^3-3xy^2=2005\Rightarrow\left[\left(\frac{x}{y}\right)^3-3\left(\frac{x}{y}\right)=\frac{2005}{y^3}\right]\times 2004\ \ldots(1)$

$y^3-3x^2y=2004\Rightarrow\left[1-3\left(\frac{x}{y}\right)^2=\frac{2004}{y^3}\right]\times 2005\ \ldots(2)$

subtract (1)& (2) & put $\frac{x}{y}=t$

$2004t^3+6015t^2-6012t-2005=0$

$\frac{y_1\cdot y_2\cdot y_3}{(y_1-x_1)(y_2-x_2)(y_3-y_3)}=\frac{1}{(1-t_1)(1-t_2)(1-t_3)}$

$=\frac{1}{1+(t_1t_2+t_2t_3+t_3t_1)-(t_1+t_2+t_3)-t_1t_2t_3}$

put values

$=1002$

put values $t_1+t_2+t_3=-6015/2004$

$t_1 t_2 + t_2 t_3 + t_3 t_1 = -6012/2004$

$t_1 t_2 t_3 = 2005/2004$

we get, 1002

so, $1002/501 = 2$